

Direction towards the left [A1] (accept negative sign if student states rightward as convention).

6 (a)

The photoemission of electrons from the zinc plate is due to the fact that the electromagnetic radiation incident on the plate exhibit a **particle nature** as opposed to wave nature. They propagate as **photons**.

Electrons are emitted from the zinc plate only when the photons each has energy that **exceed the work function** (minimum energy for photoelectron emission) of the metal through a **one-to-one interaction** between the photons and electrons.

Electrons are emitted from the metal surface because the energy of the uv photon **exceed the work function** of the metal

or

the frequency of the uv radiation (ultraviolet rays) **exceed the threshold frequency** (minimum frequency for photoelectron emission),

The emission of electrons is **independent of the intensity** as changing the intensity merely changes the rate of the incident photons and it does not alter their energy. Thus, a weak intensity uv radiation can cause photoelectron emission but an intense visible light may not cause photoelectron emission.

6 (b)

$$\text{Work function } \phi = \frac{hc}{\lambda_0} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{2.9 \times 10^{-7}}$$

$$= 6.86 \times 10^{-19} \text{ J} \quad [\text{C1}]$$

Energy of photon $E = 5.1 \text{ eV}$

$$\begin{aligned} &= (5.1)(1.60 \times 10^{-19}) \\ &= 8.16 \times 10^{-19} \text{ J} \quad [\text{C1}] \end{aligned}$$

$$E = \phi + \text{Max KE}$$

$$\text{Max KE} = E - \phi$$

$$\begin{aligned} &= (8.16 - 6.86) \times 10^{-19} \\ &= \mathbf{1.3 \times 10^{-19} \text{ J}} \quad [\text{A1}] \end{aligned}$$

6 (c)(i)

Because the electrode C is connected to the negative terminal, the electrons are **repelled** by it [B1] (ie the electric force acts opposite to the initial electron flow).

As V increases, fewer electrons per unit time reach C and hence the current I decreases. [B1]

2018 JC2 H2 Physics Prelims Exam Solution (**9749** Paper 2)

6 (c)(ii)

The electrons are emitted with a **range of KE**.

When the **opposing potential** is raised to the point (**stopping potential**) where the **current is just zero**, it is an indication that the most energetic electron is stopped on its track just short of touching C. [B1]

By COE,

Loss in Max KE = Gain in EPE

$$= (\text{Stopping potential})(\text{charge})$$

$$= V e$$

$$= \mathbf{2.2 \text{ eV}}$$

$$= \mathbf{3.36 \times 10^{-19} \text{ J}} \text{ (max KE of e) [B1]}$$

7 (a)